

Industry adoption of open source middleware in IoT

The market for IoT middleware was valued at US\$ 6.44 billion in 2018 and is expected to reach a value of US\$ 18.68 billion by 2024 at a CAGR of 19.72 per cent, over the forecast period 2019-2024 (<https://www.mordorintelligence.com/industry-reports/iot-middle-ware-market>).

According to an Ericsson forecast (<https://www.ericsson.com/en/mobility-report/internet-of-things-forecast>), there will be around 29 billion connected devices in use by 2022, of which around 18 billion will be related to IoT. Gartner forecasts that 14.2 billion connected things will be in use in 2019, and that the total will reach 25 billion by 2021, producing immense volume of data.

IoT middleware and its features

Middleware acts as an agent between the service providers (IoT devices) and service consumers (enterprise applications). It is a software layer that sits in between applications and objects. It is a mediator interface that enables the interaction between the Internet and 'things'. It hides the heterogeneity among the devices, components and technology of an IoT system. Middleware provides solutions to frequently encountered problems, such as interoperability, security and dependability. The following are the important features of middleware, which improve the performance of devices.

Flexibility: This feature helps in establishing better connectivity, which improves the communication between applications and things. There are different kinds of flexibility (e.g., response time, faster to evolve and change).

Transparency: Middleware hides many complexities and architectural information details from both the application and the object sides, so that the two can communicate with minimum knowledge of either side.

Interoperability: This functionality allows two sets of applications on

interconnected networks to exchange data and services meaningfully with different assumptions on protocols, data models, and configurations.

Platform portability: An IoT platform should be able to communicate from everywhere, anytime with any device. Middleware runs on the user side and can provide independence from network protocols, programming languages, OSs and others.

Re-usability: This feature makes designing and developing easier by modifying system components and assets for specific requirements, which results in cost efficiency.

Maintainability: Maintainability has a fault tolerance approximation. Middleware performs maintainability efficiently and extends the network.

Security: Middleware should provide different security measures for ubiquitous applications and pervasive environments. Authentication, authorisation and access control helps in verification and accountability.

Characteristics of open source IoT middleware

An open source IoT middleware platform should be fault-tolerant and highly available. It has the following characteristics:

- No vendor lock-in, and it comes with the surety of seamless integration of enterprise-wide tools, applications, products and systems developed and deployed by different organisations and vendors.
- Open source middleware increases the productivity, speeds up time to market, reduces risk and increases quality.
- Adoption of open source middleware enhances the interoperability with other enterprise applications because of the ability to reuse recommended software stacks, libraries and components.
- IoT middleware platforms should support open APIs, deployment models of the cloud, and be highly available.

- It should support open data formats like RestAPI, JSON, XML and Java, and be freely available
- An IoT middleware platform should support multi-service and heterogeneous devices, and be compatible with the hardware for sensing environmental information.
- Migration to any new platform or system should be seamless. It should be possible to adopt or integrate with any solution.
- The information data model should be distributed and extensible, providing availability and scalability to the system.
- An IoT middleware platform should support major communication protocols like MQTT, CoAP, HTTP, WebSockets, etc.
- An IoT middleware platform should support different security features like encryption, authentication, authorisation and auditing.
- It should support technologies such as M2M applications, real-time analytics, machine learning, artificial intelligence, analytics, visualisation and event reporting.

IoT middleware architecture

The middleware mediates between IoT data producers and the consumers. APIs for interactions with the middleware are based on standard application protocols.

API endpoints for accessing the data and services should be searchable via an open catalogue, and should contain linked metadata about the resources.

The device manager communicates messages to the devices. The database needs to access and deliver messages to the devices with minimum latency.

Data processing involves data translation, aggregation and data filtering on the incoming data, which enables real-time decision making at the edge. The database needs to support high-speed reads and writes with sub-millisecond latency. It helps in performing complex analytical computations on the data.